



Course Name:	Differential Equations	Course Code:	MT1006
Degree Program:	BS(CS) BS(AI)	Semester:	Fall 2023
Exam Duration:	180 Minutes	Total Marks:	85
Paper Date:	Saturday, December 23, 2023	Obtained Marks	
Sections:	ALL	No of Page(s):	09
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Student Name: _____ Roll No. _____ Section: _____

Instruction/Notes: Attempt all questions. Programmable calculators are not allowed.

	Q-1	Q-2	Q-3	Q-4	Q-5	Q-6	Q-7	Q-8	Total
Total Marks	10	10	10	10	10	10	10	15	85
Obtained Marks									

Question # 1

10 Points

Identify the type of differential equation and solve the given 1st order initial value problem

$$(1+x^2) \frac{dy}{dx} + 2xy = f(x), y(0) = 0 \text{ where } f(x) = \begin{cases} x, & 0 \leq x < 1 \\ -x, & x \geq 1. \end{cases}$$

we solve the DE for $y(x)$ first on the interval $[0, 1)$ and then on the interval $[1, \infty)$.

$$(1+x^2) \frac{dy}{dx} + 2xy = x$$

$$\text{or } \frac{dy}{dx} + \frac{2x}{1+x^2} \cdot y = \frac{x}{1+x^2} \quad \text{--- (A)}$$

$$\text{here } p(x) = \frac{2x}{1+x^2}$$

$$I.F = e^{\int p(x) dx} = e^{\int \frac{2x}{1+x^2} dx} = e^{\ln|1+x^2|} = 1+x^2$$

Multiply eq (A) by $I.F = 1+x^2$

$$(1+x^2) \frac{dy}{dx} + 2xy = x$$

$$\frac{d}{dx} [(1+x^2) y] = x$$

$$\text{or } (1+x^2) y = \int x dx$$

$$(1+x^2) y = \frac{x^2}{2} + C_1$$

or

$$y = \frac{1}{1+x^2} \left(\frac{x^2}{2} + C_1 \right)$$

02

$$y(0) = 0$$

$$\Rightarrow \boxed{y = \frac{x^2}{2(x^2+1)}} \Rightarrow \boxed{C_1 = 0}$$

01

And for $[1, \infty)$

$$(1+x^2) \frac{dy}{dx} + 2xy = -x$$

Similarly as in 1st case.

or

$$\frac{d}{dx} [(1+x^2) \cdot y] = -x$$

or

$$y = \frac{1}{1+x^2} \left(-\frac{x^2}{2} + C_2 \right)$$

03

Hence

$$y(x) = \begin{cases} \frac{x^2}{2(x^2+1)} & ; 0 \leq x < 1 \\ \frac{1}{1+x^2} \left(-\frac{x^2}{2} + C_2 \right) & ; x \geq 1 \end{cases}$$

02

A thermometer is removed from a room where the temperature is 70°F and is taken outside, where the air temperature is 10°F . After one-half minute the thermometer reads 50°F . What is the reading of thermometer after one minute? How long will it take for the thermometer to reach 18°F ?

$$\frac{dT}{dt} = k(T - T_m)$$

$$T_m = 10^\circ\text{F}$$

$$T(0) = 70^\circ\text{F}$$

$$\& T\left(\frac{1}{2}\right) = 50^\circ\text{F}$$

$$\frac{dT}{dt} = k(T - 10)$$

$$\frac{1}{T-10} dT = k dt$$

Integrating both sides, we get

$$\int \frac{1}{T-10} dT = \int k dt$$

$$\ln|T-10| = kt + C_1$$

$$\text{or } T-10 = e^{kt+C_1}$$

$$\text{or } \boxed{T(t) = 10 + ce^{kt}} \quad \text{--- (A) --- (0.3)}$$

$$T(0) = 70$$

$$\Rightarrow 70 = 10 + ce^0 \Rightarrow \boxed{c = 60} \quad \text{--- (0.5)}$$

$$\text{(A)} \Rightarrow \boxed{T(t) = 10 + 60e^{kt}} \quad \text{--- (B)}$$

$$T\left(\frac{1}{2}\right) = 50$$

$$\Rightarrow \text{(B)} \Rightarrow 50 = 10 + 60e^{\frac{1}{2}k}$$

$$\Rightarrow \boxed{k = 2 \ln\left(\frac{2}{3}\right)}$$

$$\text{(0.5)}$$

So,

$$T(t) = 10 + 60 e^{2 \ln(\frac{2}{3}) t}$$

(a)

$$T(1) = 10 + 60 e^{2 \ln(\frac{2}{3}) \cdot 1}$$

02

$$\Rightarrow T(1) = 36.67^\circ$$

(b)

$$\text{when } T(t) = 18^\circ, \quad t = ?$$

$$T(t) = 10 + 60 e^{2 \ln(\frac{2}{3}) t}$$

02

$$18 = 10 + 60 e^{2 \ln(\frac{2}{3}) t}$$

$$\Rightarrow t \approx 2.5 \text{ minutes}$$

- a) Given that $y_1 = x^2 \cos(\ln x)$ is a solution of the differential equation $x^2 y'' - 3xy' + 5y = 0$. Use reduction of order to find a second solution y_2 .
- b) Determine whether the two solutions y_1 and y_2 of $x^2 y'' - 3xy' + 5y = 0$ form a fundamental set of solutions? Justify your answer.

Sol:- $x^2 y'' - 3xy' + 5y = 0 \Rightarrow y'' - \frac{3}{x} y' + \frac{5}{x^2} y = 0$

Since, $P(x) = -\frac{3}{x}$

$$y_2 = y_1(x) \int \frac{e^{-\int P(x) dx}}{(y_1(x))^2} dx$$

$$\begin{aligned} y_2 &= x^2 \cos(\ln x) \int \frac{e^{3 \int \frac{1}{x} dx}}{x^4 \cos^2(\ln x)} dx \\ &= x^2 \cos(\ln x) \int \frac{e^{3 \ln x}}{x^4 \cos^2(\ln x)} dx \\ &= x^2 \cos(\ln x) \int \frac{e^{\ln x^3}}{x^4 \cos^2(\ln x)} dx \\ &= x^2 \cos(\ln x) \int \frac{x^3}{x^4 \cos^2(\ln x)} dx \\ &= x^2 \cos(\ln x) \int \frac{1}{x \cos^2(\ln x)} dx \end{aligned}$$

Let $u = \ln x$
 $du = \frac{1}{x} dx$

$$\begin{aligned} &= x^2 \cos(\ln x) \int \frac{1}{\cos^2 u} du \\ &= x^2 \cos(\ln x) \int \sec^2 u du \\ &= x^2 \cos(\ln x) (\tan u) \\ &= x^2 \cos(\ln x) \tan(\ln x) \\ y_2 &= x^2 \cos(\ln x) \cdot \frac{\sin(\ln x)}{\cos(\ln x)} = x^2 \sin(\ln x) \end{aligned}$$

$$y_2 = x^2 \sin(\ln x),$$

b) Since $y_1 = x^2 \cos(\ln x)$, $y_2 = x^2 \sin(\ln x)$
 y_1 & y_2 make fundamental sol.
 If y_1 & y_2 are LI

$$\begin{aligned} \Rightarrow W &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} x^2 \cos(\ln x) & x^2 \sin(\ln x) \\ -x^2 \sin(\ln x) + 2x \cos(\ln x) & 2x \sin(\ln x) + x^2 \cos(\ln x) \end{vmatrix} \\ &= 2x^3 \cos(\ln x) \sin(\ln x) + x^4 \cos^2(\ln x) \\ &\quad - \left(x^4 \sin^2(\ln x) + 2x^3 \sin(\ln x) \cos(\ln x) \right) \\ &= 2x^3 \cos(\ln x) \sin(\ln x) + x^4 \cos^2(\ln x) + x^4 \sin^2(\ln x) \\ &\quad - 2x^3 \sin(\ln x) \cos(\ln x) \\ &= x^4 (\cos^2(\ln x) + \sin^2(\ln x)) \\ &= x^4 \neq 0 \end{aligned}$$

$\Rightarrow y_1$ & y_2 form a fundamental solution.

Find the solution of given 3rd order initial value problem by method of undetermined coefficients.

$$y''' - 2y'' - y' = x^2 + 5, y(0) = 2, y'(0) = 2, y''(0) = -1$$

$$D^3 - 2D^2 + D = 0$$

$$D(D^2 - 2D + 1) = 0$$

$$D = 0; D^2 - 2D + 1 = 0$$

$$(D - 1)^2 = 0$$

$$D = 1, 1$$

$$y_c = C_1 e^x + C_2 x e^x + C_3 \quad (1)$$

Let $y_p = (Ax + B)e^x + C$
 $= (Ax^2 + Bx)e^x + Cx$

$$y_p = (Ax^3 + Bx^2)e^x + Cx^2 \quad (2)$$

$$y_p = y_{p1} + y_{p2} \quad (3)$$

$$y_{p1} = (Ax^3 + Bx^2)e^x$$

$$y_{p1}' = (3Ax^2 + 2Bx)e^x + (Ax^3 + Bx^2)e^x$$

$$y_{p1}'' = (6Ax + 2B)e^x + 2(3Ax^2 + 2Bx)e^x + (Ax^3 + Bx^2)e^x$$

$$y_{p1}''' = 6Ae^x + 3(6Ax + 2B)e^x + 3(3Ax^2 + 2Bx)e^x + (Ax^3 + Bx^2)e^x$$

Comparing coefficients

Put in (1)

$$x^3 e^x: 6A - 12A - 4B + 2B = 1$$

$$x^2 e^x: -6A - 2B = 1$$

$$x e^x: 6A - 2B = 0$$

$$\left. \begin{aligned} A &= -1/12 \\ B &= -1/4 \end{aligned} \right\} (2)$$

$$y_p = \left(-\frac{1}{12}x^3 - \frac{1}{4}x^2\right)e^x - 5x$$

General sol

$$y = y_c + y_p$$

$$y = c_1 e^x + c_2 x e^x + c_3 - \left(\frac{1}{12}x^3 - \frac{1}{4}x^2\right)e^x - 5x$$

$$y' = c_1 e^x + c_2 e^x + \frac{1}{4}x^2 e^x + \frac{1}{2}x e^x - \frac{1}{12}x^3 e^x + \frac{1}{4}x^2 e^x - 5$$

$$y'' = c_1 e^x + c_2 e^x - \frac{1}{2}x e^x - \frac{1}{4}x^2 e^x + \frac{1}{2}e^x + \frac{1}{2}x e^x - \frac{1}{6}x^2 e^x - \frac{1}{12}x^2 e^x + \frac{1}{2}x^2 e^x + \frac{1}{4}x^2 e^x - 5$$

Applying I.C.S

$$y(0) = 2$$

$$c_1 + c_3 = 2$$

$$y'(0) = 2$$

$$c_1 + c_2 = 2$$

$$y''(0) = -1$$

$$c_1 + 2c_2 = -6$$

$$\Rightarrow c_1 = 10, c_2 = -8, c_3 = 8$$

$\Rightarrow$

$$y = 10e^x - 8xe^x + 8 - \frac{1}{12}x^3 e^x - \frac{1}{4}x^2 e^x - 5x$$

①

Find the general solution of $y'' + 2y' + y = e^{-t} \ln t$ by variation of parameters.

$$D^2 + 2D + 1 = 0$$

$$(D+1)^2 = 0$$

$$D = -1, -1$$

$$\Rightarrow \boxed{y_c = c_1 e^{-t} + c_2 t e^{-t}}$$

(2)

$$y_p = u(t) y_1 + v(t) y_2$$

(1)

Let

$$y_1 = e^{-t}$$

$$y_2 = t e^{-t}$$

$$y_1' = -e^{-t}$$

$$y_2' = e^{-t} - t e^{-t}$$

(1)

$$W = \begin{vmatrix} e^{-t} & t e^{-t} \\ -e^{-t} & e^{-t}(1-t) \end{vmatrix} = e^{-t} \cdot e^{-t} \begin{vmatrix} 1 & t \\ -1 & 1-t \end{vmatrix} = e^{-2t} (1 - t + t) = e^{-2t}$$

$$\boxed{W = e^{-2t}}$$

(1)

$$\boxed{r(t) = e^{-t} \ln t}$$

$$u(t) = - \int \frac{y_2 r(t)}{W} dt = - \int \frac{t e^{-t} e^{-t} \ln t}{e^{-2t}} dt$$

$$= - \int t \ln t dt \Rightarrow \boxed{u(t) = -\frac{t^2}{2} \ln t + \frac{1}{4} t^2}$$

$$v(t) = \int \frac{y_1 r(t)}{W} dt =$$

$$\boxed{v(t) = t \ln t - t}$$

(7)

$$\boxed{y = c_1 e^{-t} + c_2 t e^{-t} + \left(\frac{1}{4} t^2 - \frac{t^2}{2} \ln t \right) e^{-t} + (t \ln t - t) t e^{-t}}$$

Find a particular solution of the Cauchy-Euler equation $x^2 y'' - 4xy' + 6y = \ln x^2$ considering the conditions $y(1) = 1, y'(1) = 0$.

Let $y = x^m$

$$f(x) = \frac{\ln x^2}{x^2} = \frac{2 \ln x}{x^2}$$

$$m(m-1) - 4m + 6 = 0$$

$$m^2 - 3m - 2m + 6 = 0$$

$$m = 2; m = 3$$

$$y_c = C_1 y_1 + C_2 y_2 = C_1 x^2 + C_2 x^3$$

$$y_1 = x^2 \quad (1) \quad y_2 = x^3 \quad (2)$$

$$W = \begin{vmatrix} x^2 & x^3 \\ 2x & 3x^2 \end{vmatrix} = +x^4 \quad (3)$$

$$u_1 = - \int \frac{y_2 f(x)}{W} dx$$

$$u_1 = - \int \frac{2x^3 \frac{\ln x}{x^2}}{x^4} dx$$

$$u_1 = -2 \int \frac{\ln x}{x^3} dx$$

$$= -2 \ln x \frac{x^{-2}}{-2} + 2 \int \frac{1}{x} \frac{x^{-2}}{-2} dx$$

$$= -\frac{2}{2} \ln x \cdot \frac{1}{x} - \frac{2}{2} \int \frac{1}{x^3} dx$$

$$= + \frac{1}{x^2} \ln x - \frac{1}{2} \frac{x^{-2}}{-2}$$

$$+ \frac{1}{x^2} \ln x + \frac{1}{2x^2}$$

$$u_2 = \int \frac{y_1 f(x)}{W} dx$$

$$u_2 = \int \frac{x^2 \frac{2 \ln x}{x^2}}{x^4} dx$$

$$u_2 = +2 \int \frac{\ln x}{x^4} dx$$

$$= +2 \ln x \frac{x^{-3}}{-3} + 2 \int \frac{x^{-3}}{-3} \frac{1}{x} dx$$

$$= -\frac{2}{3} \ln x + \frac{2}{3} \int \frac{1}{x^4} dx$$

$$= -\frac{2}{3} \ln x - \frac{2}{3} \left(\frac{x^{-3}}{-3} \right)$$

$$= -\frac{2}{3} \ln x + \frac{2}{9x^3}$$

$$y_p = y_1 + y_2$$

$$x^2 \left(\frac{1}{x^2} \ln x + \frac{1}{2x^2} \right) + \left(\frac{-2}{3x^3} \ln x - \frac{2}{9x^3} \right) x^3$$

$$= + \ln x + \frac{1}{2} - \frac{2}{3} \ln x - \frac{2}{9}$$

$$= + \frac{1}{3} \ln x + \frac{5}{18}$$

$$y = y_c + y_p = C_1 x^2 + C_2 x^3 + \frac{1}{3} \ln x + \frac{5}{18} \quad \text{--- (a)}$$

$$y' = 2C_1 x + 3C_2 x^2 + \frac{1}{3x} + 0 \quad \text{--- (b)}$$

$$\text{(a)} \Rightarrow y(1) = 1$$

$$1 = C_1 + C_2 + 5/18$$

$$C_1 + C_2 = 1 - 5/18$$

$$C_1 + C_2 = \frac{13}{18} \quad \text{--- (c)}$$

$$\text{(b)} \Rightarrow y'(1) = 0$$

$$0 = 2C_1 + 3C_2 + \frac{1}{3}$$

$$2C_1 + 3C_2 = -\frac{1}{3} \quad \text{--- (d)}$$

$$C_1 = \quad ; C_2 = 0$$

$$y = x^2 + x^3 + \frac{1}{3} \ln x + \frac{5}{18}$$

Use separation of variables to find, if possible, product solution for the 2nd order linear partial differential equation

$$\frac{\partial^2 u}{\partial^2 x} - u = \frac{\partial u}{\partial y} \quad \text{--- (1)}$$

let

$$u = X(x) Y(y) \quad \text{--- (2)}$$

$$\frac{\partial u}{\partial x} = X' Y$$

$$\frac{\partial^2 u}{\partial x^2} = X'' Y$$

$$\frac{\partial u}{\partial y} = X Y'$$

By substitution in (1)

$$4 X'' Y - X Y = X Y'$$

÷ by $4XY$

$$\frac{X''}{X} - \frac{1}{4} = \frac{1}{4} \frac{Y'}{Y}$$

$$\frac{X''}{X} = \frac{1}{4} \frac{Y'}{Y} + \frac{1}{4}$$

Now

$$\frac{X''}{X} = -\lambda, \quad \frac{1}{4} \frac{Y'}{Y} + \frac{1}{4} = -\lambda$$

let

a) $\lambda = 0$

$$\int X'' = 0$$

$$X = c_1 x + m_1$$

$$\frac{1}{4} \frac{Y'}{Y} + \frac{1}{4} = 0$$

$$\int \frac{Y'}{Y} = \int -1 dy$$

$$\ln Y = -y + c$$

$$Y = c_2 e^{-y}$$

b) $\lambda = \alpha^2$

$$0 = X'' + \alpha^2 X$$

$$X = a_1 \cos \alpha x + b_1 \sin \alpha x$$

and

$$\int \frac{Y'}{Y} = \int (-4\alpha^2 - 1) dy$$

$$\ln Y = -(4\alpha^2 + 1)y + a$$

$$e^{\ln Y} = e^{-(4\alpha^2 + 1)y + a}$$

$$Y = a_2 e^{-(4\alpha^2 + 1)y}$$

$$u = (a_1 \cos \alpha x + b_1 \sin \alpha x) (a_2 e^{-(4\alpha^2 + 1)y})$$

$$u = A_2 e^{-(4\alpha^2 + 1)y} \cos \alpha x + B_2 e^{-(4\alpha^2 + 1)y} \sin \alpha x$$

c) $\lambda = -\alpha^2$

$$0 = X'' - \alpha^2 X$$

$$X = a_2 \cosh \alpha x + b_2 \sinh \alpha x$$

$$\int \frac{Y'}{Y} = \int (4\alpha^2 - 1) dy$$

$$\ln Y = (4\alpha^2 - 1)y + b$$

$$Y = b_2 e^{(4\alpha^2 - 1)y}$$

$$u = (a_2 \cosh \alpha x + b_2 \sinh \alpha x) (b_2 e^{(4\alpha^2 - 1)y})$$

$$u = A_3 e^{(4\alpha^2 - 1)y} \cosh \alpha x + B_3 e^{(4\alpha^2 - 1)y} \sinh \alpha x$$

Solve the given system of ordinary differential equations by systematic elimination.

$$x' - 4x + y'' = t^2$$

$$x' + x + y' = 0$$

Sol:- First we will write system of DE in differential operator notation

$$(D-4)x + D^2y = t^2 \quad \text{--- (1) } \times \text{ by } (D+1)$$

$$(D+1)x + Dy = 0 \quad \text{--- (2) } \times \text{ by } (D-4)$$

$$(D+1)(D-4)x + D^2(D+1)y = (D+1)t^2$$

$$- (D+1)(D-4)x + D(D-4)y = 0$$

$$D^2(D+1)y - D(D-4)y = (D+1)t^2$$

$$(D^3 + 4D)y = t^2 + 2t \quad \text{--- (3)}$$

Auxiliary eq of eq (3)

$$m^3 + 4m = 0$$

$$m(m^2 + 4) = 0$$

$$m_1 = 0, \quad m_2 = 2i, \quad m_3 = -2i$$

$$y_c = C_1 + C_2 \cos 2t + C_3 \sin 2t \quad \text{--- (4)}$$

For particular solution let

(VC Method)

$$y_p = At^3 + Bt^2 + Ct$$

$$y_p' = 3At^2 + 2Bt + C$$

$$y_p'' = 6At + 2B$$

$$y_p''' = 6A$$

using in (4)

$$12At^2 + 8Bt + 6A + 4C = t^2 + 2t$$

By comparing coefficients

$$12A = 1 \Rightarrow A = \frac{1}{12}$$

$$8B = 2 \Rightarrow B = \frac{1}{4}$$

$$6A + 4C = 0 \Rightarrow C = -\frac{1}{8}$$

$$y_p = \frac{1}{12}t^3 + \frac{1}{4}t^2 - \frac{1}{8}t$$

$$y = y_c + y_p$$

$$y(t) = C_1 + C_2 \cos 2t + C_3 \sin 2t + \frac{1}{12}t^3 + \frac{1}{4}t^2 - \frac{1}{8}t$$

Now for, $y(t)$.

$$(D-4)y + D^2 y = t^2$$

or eq (2) by D.

$$-D(D+1)y + D^2 y = 0$$

$$\frac{[(D-4) - D^2 - D]y = t^2}{(D^2 + 4)y = -t^2} \text{ --- (5)}$$

$$m^2 + 4 = 0$$

$$m_1 = 2i, m_2 = -2i$$

$$y_c = C_4 \cos 2t + C_5 \sin 2t$$

using in eq (6)

$$y_p = -\frac{1}{4}t^2 + \frac{1}{8}$$

Now let

$$x_p = At^2 + Bt + C \text{ --- (6)}$$

$$x_p' = 2At + B$$

$$x_p'' = 2A$$

using in eq (5)

$$2A + 4At^2 + 4Bt + 4C = -t^2$$

$$4A = -1 \Rightarrow A = -\frac{1}{4}$$

$$4B = 0 \Rightarrow B = 0$$

$$2A + 4C = 0 \Rightarrow C = \frac{1}{8}$$

$$x(t) = C_4 \cos 2t + C_5 \sin 2t - \frac{1}{4}t^2 + \frac{1}{8}$$

now to express C_4 and C_5 in terms of C_2 and C_3

$$x' + x + y' = 0$$

$$-2C_4 \sin 2t + 2C_5 \cos 2t - \frac{1}{2}t + C_4 \cos 2t + C_5 \sin 2t - \frac{1}{4}t^2 + \frac{1}{8} - 2C_2 \sin 2t + 2C_3 \cos 2t + \frac{1}{4}t^2 + \frac{1}{2}t - \frac{1}{8}$$

$$(C_5 - 2C_4 - 2C_2) \sin 2t + (2C_5 + C_4 + 2C_3) \cos 2t = 0 \quad (7)$$

$$\begin{aligned} C_5 - 2C_4 - 2C_2 &= 0 \\ 2C_5 + C_4 + 2C_3 &= 0 \\ \text{②} \rightarrow 2C_5 - 4C_4 - 4C_2 &= 0 \\ -2C_5 + C_4 + 2C_3 &= 0 \end{aligned}$$

$$-5C_4 - 4C_2 - 2C_3 = 0$$

$$C_4 = -\frac{1}{5}(4C_2 + 2C_3)$$

$$C_5 - 2C_4 - 2C_2 = 0$$

$$4C_5 - 2C_4 + 4C_3 = 0$$

$$5C_5 - 2C_2 + 4C_3 = 0$$

$$C_5 = \frac{1}{5}(2C_2 - 4C_3)$$

$$x(t) = -\frac{1}{5}(4C_2 + 2C_3) \cos 2t + \frac{1}{5}(2C_2 - 4C_3) \sin 2t - \frac{1}{4}t^2 + \frac{1}{8}$$

$$y(t) = C_1 + C_2 \cos 2t + C_3 \sin 2t + \frac{1}{12}t^3 + \frac{1}{4}t^2 + \frac{1}{8}$$